IAC TECHNOLOGY DIVISION

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MUSOLSONG YAMLEDITOR SOFTWARE SYSTEM - USER MANUAL

PROJECT / DESTINATION:

MUSOL - ESPECTRO POLARÍMETRO SOLAR

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1.0	18-03-2026	Initial Issue

SUMMARY

This document describes the use of the MUSOLSONG YAMLEditor software. A graphical tool for creating and editing YAML configuration files specifically designed for use with the MUSOLSONG Controller application.

LIST OF ABBREVIATIONS

Abbreviation	Description
AD	Applicable Document
IAC	Instituto de Astrofísica de Canarias
RD	Referenced Document
TBC	To Be Confirmed
TBD	To Be Defined

TABLE OF CONTENTS

AUTHORS LIST **2**

APPROVAL CONTROL **2**

DOCUMENT CHANGES RECORD **2**

SUMMARY **3**

THIS DOCUMENT DESCRIBES THE USE OF THE MUSOLSONG YAMLEDITOR SOFTWARE. A GRAPHICAL TOOL FOR CREATING AND EDITING YAML CONFIGURATION FILES SPECIFICALLY DESIGNED FOR USE WITH THE MUSOLSONG CONTROLLER APPLICATION **3**

LIST OF ABBREVIATIONS **4**

1. INTRODUCTION **6**

1.1. BACKGROUND 6

2. OVERVIEW **6**

2.1. SYSTEM STRUCTURE 6

2.2. SYSTEM REQUIREMENTS 6

2.3. INSTALLATION 7

2.4. GETTING STARTED 7

2.5. INTERFACE OVERVIEW 7

2.5.1. Menu Bar 7

2.5.2. Main Tabs 7

2.5.3. Control Buttons 9

2.6. USING THE APPLICATION 9

2.6.1. Step 1: Select Mode 9

2.6.2. Step 2: Configure Selected Mode 10

2.6.3. Step 3: Validate and Save 10

2.7. IMPORTANT RESTRICTIONS FOR MUSOLSONG 10

2.8. VISUAL GUIDE TO INTERFACE ELEMENTS 10

2.8.1. Calibration Tab Elements: 10

2.8.2. Observation Tab Elements: 11

2.9. TROUBLESHOOTING 11

2.9.1. Common Issues 11

2.10. EXAMPLE OF YAML CONFIGURATION FILES 11

2.10.1. Calibration Mode Example 11

2.10.2. Observation Mode Example 11

1. INTRODUCTION

1.1. Background

MUSOLSONG is part of a research project aimed at upgrading the Solar-SONG instrument, which operates on the Hertzsprung-SONG telescope at the Teide Observatory. The main objective of this project is to study the Sun's global and cyclical magnetism, complemented by investigations of its interior to better understand the origin of our star's cyclical activity.

The project involves the design, manufacture, and integration of the MUSOL polarimeter, its coupling with the SONG spectrograph, and the provision of all the necessary infrastructure to ensure the proper operation of the polarimeter.

2. OVERVIEW

MUSOLSONG is a software project containing two software applications: The musolsong-controller and the musolsong-yamleditor.

The MUSOLSONG YAMLEditor is a Python-based graphical application for creating and editing YAML configuration files specifically designed for use with the MUSOLSONG Controller application. It allows you to configure calibration and observation modes with precise modulation parameters.

Important Note: While this application can generate files containing both modes(calibration/observation), the MUSOLSONG Controller application requires that you **do not mix modes in one file**. You **must enable either** "Calibration Mode" OR "Observation Mode" - never both simultaneously.

2.1. System Structure

- **Controller (GUI)**: Centralized application for operations management
- **YAML Editor**: Application used to create/edit configuration files for the controller

2.2. System Requirements

Operating System: The application was developed on Linux OS. However, it should be compatible with other systems (Windows or mac Os) supporting Python and PyQt6

Required Python Version:

Python 3.6 or higher

Required Packages:

PyQt6 >=6.91,

PyYAML>=5.4.1

2.3. Installation

1. Ensure Python is installed on your computer
2. Install required libraries:

   pip install PyQt6 PyYAML
3. Install the musolsong package

   pip install musolsong-2026.0.1.3-py3-none-any.whl

2.4. Getting Started

1. `Start the application with:`

   `musolsong-yamleditor`
2. `The main window will open with two tabs: "Calibration" and "Observation"`

2.5. Interface Overview

2.5.1. `Menu Bar`

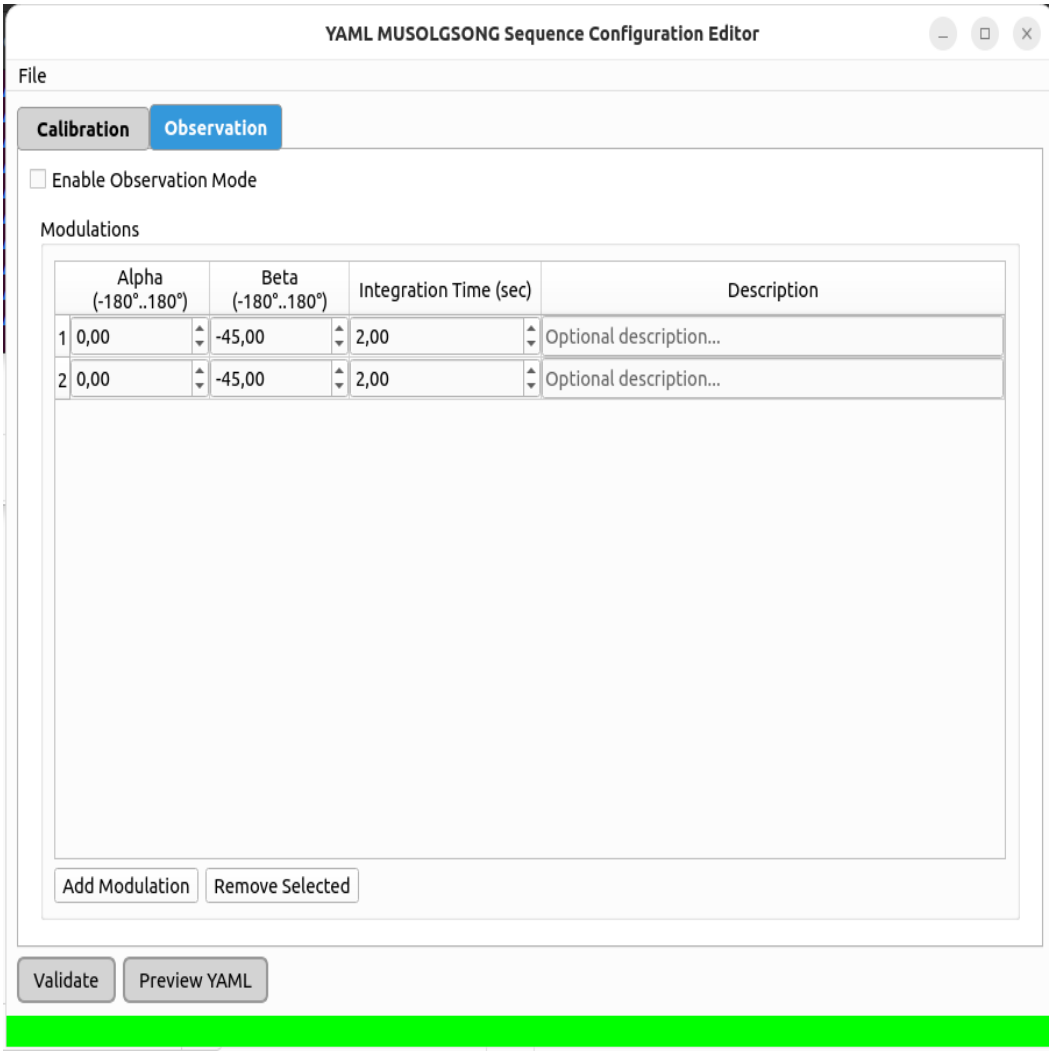
- `File` → New: Create a new configuration
- `File` → Open: Open an existing YAML file
- `File` → Save: Save the current configuration
- `File` → Save As: Save with a new filename
- `File` → Exit: Close the application

2.5.2. `Main Tabs`

1. **`Calibration Tab:`** Configure calibration mode parameters

YAML MUSOLGSONG Sequence Configuration Editor
File
Calibration
Observation
Enable Calibration Mode
Retarding Plate Offsets
Incremental angle per cycle:
5 degrees
Valid values:
Must be a factor of 360 (e.g., 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 18, 20, 24, 30, 36, 40, 45, 60, 72, 90, 120, 180, 360)
Number of cycles: 72
Modulations
Alpha (-180°..180°)
Beta (-180°..180°)
Integration Time (sec)
Description
1 0,00 -45,00 2,00 Optional description...
2 0,00 -45,00 2,00 Optional description...
Add Modulation
Remove Selected
Validate
Preview YAML

2. **`Observation Tab:`** Configure observation mode parameters



2.5.3. Control Buttons

- **`Validate:`** Check your configuration for errors. At least one mode should be selected (must check either "Enable Calibration Mode" OR "Enable Observation Mode"
- **`Preview YAML:`** Validates and shows the generated YAML before saving. At least one mode should be selected (must check either "Enable Calibration Mode" OR "Enable Observation Mode"

2.6. Using the Application

2.6.1. Step 1: Select Mode

- `Important:` Select ONLY ONE mode for MUSOLSONG compatibility
- `Check either` "Enable Calibration Mode" OR "Enable Observation Mode"

- `Never` enable both modes simultaneously

2.6.2. Step 2: Configure Selected Mode

`For Calibration Mode:`

1. `Set the Retarding Plate Offset` (must be a factor of 360°)
    - Valid values: 1, 2, 3, 4, 5, 6, 8, 9, 10, 12, 15, 18, 20, 24, 30, 36, 40, 45, 60, 72, 90, 120, 180, 360
    - The application will automatically calculate and display the required number of cycles
2. `Add modulation entries using the table:`
    - `Alpha:` Angle between -180° and 180° (first column)
    - `Beta:` Angle between -180° and 180° (second column, default: -45°)
    - `Integration Time:` Between 0.1 and 300 seconds (third column, default: 2.0s)
    - `Description:` Optional text description (fourth column)
3. `Use "Add Modulation"` button to add new rows
4. `Use "Remove Selected"` button to delete selected rows
5. `Click on any cell to edit` its value

For Observation Mode:

1. `Add modulation entries using the table` (same parameters as calibration)
    - `Note:` Observation mode does not require a retarding plate offset
2. `Use "Add Modulation" and "Remove Selected"` buttons to manage rows

2.6.3. Step 3: Validate and Save

1. `Click Validate` to check for errors (highly recommended)
2. `Click Preview YAML` to see the generated configuration before saving
3. `Use Save or Save As from the File menu` to save your YAML file

2.7. Important Restrictions for MUSOLSONG

Enable ONLY ONE mode per file

For calibration: Retarding plate offset must be a factor of 360°

Each mode must have at least one modulation

NEVER enable both modes in the same file

Don't use invalid offset values in calibration mode

2.8. Visual Guide to Interface Elements

2.8.1. Calibration Tab Elements:

1. **`Enable Calibration Mode checkbox`** - Activates calibration mode

2. **`Incremental angle per cycle`** - Set your retarding plate offset (must be factor of 360)
3. **`Valid values note`** - Reminder of retarding plate offset acceptable values
4. **`Number of cycles display`** - Automatically calculated based on your offset
5. **`Modulations table`** - Where you enter alpha, beta, integration time, and descriptions
6. **`Add Modulation button`** - Adds a new row to the table
7. **`Remove Selected button`** - Deletes the currently selected row

2.8.2. `Observation Tab Elements:`

1. **`Enable Observation Mode checkbox`** - Activates observation mode
2. **`Modulations table`** - Same structure as calibration table
3. **`Control buttons`** - Same add/remove functionality as calibration tab

2.9. Troubleshooting

2.9.1. Common Issues

1. **Validation errors**:
   - Ensure only one mode is enabled
   - Check that calibration offset is a valid factor of 360
   - Ensure each enabled mode has at least one modulation
2. **File won't open**:
   - Check that the file is a valid YAML format
   - Ensure the file follows the expected structure
3. **Application won't start**:
   - Verify Python and required libraries are installed
   - Check for error messages in the terminal/console

2.10. Example of YAML Configuration files

2.10.1. Calibration Mode Example

```
modes:
  calibration:
    retarding_plate_offsets: 5
    modulations:
    - alpha: 0.0
      beta: -45.0
      SONG_int_time: 2.0
      description: First calibration point
```

2.10.2. Observation Mode Example

```
modes:
  observation:
    modulations:
    - alpha: 30.0
      beta: -45.0
      SONG_int_time: 5.0
      description: Primary observation
```